

Lesson 3: Georeferencing

Practical Session: Converting Scanned Maps into Spatially Referenced Data in QGIS

Objectives

1. Understand the concept of georeferencing and why it is necessary for scanned or historical maps.
2. Learn how Ground Control Points (GCPs) are used to assign real-world coordinates to a raster image.
3. Georeference a 1958 USGS topographic map of Mount St. Helens and export it as a spatially referenced GeoTIFF.
4. Compare the georeferenced historical map against a modern basemap to observe the dramatic topographic changes caused by the 1980 eruption.

Data

Download the 1958 Mount St. Helens 15-minute topographic map from the USGS topoView portal (<https://ngmndb.usgs.gov/topoview/>). Search for "Mount St. Helens, WA" and look for the 1958 (HTMC, 1968 ed., 1:62500 scale) data. Download the map as a JPEG and unzip it in your Lec3/Data folder. The data file should have the name: `WA_MtStHelens_242704_1958_62500_geo_jpg`

Preliminary Setup

1. Launch QGIS and open a new, empty project.
2. Set the Project CRS to EPSG:4326 (WGS 84) via the status bar in the bottom-right corner of the interface.

Step 1: Launching the Georeferencer Tool

QGIS includes a dedicated Georeferencer tool for assigning spatial coordinates to raster images.

Procedure

1. In the QGIS menu bar, navigate to Layer, then select Georeferencer. In older versions of QGIS (prior to 3.26), this tool was found under Raster, then Georeferencer.
2. The Georeferencer window is divided into two sections: a large upper panel where your image will be displayed, and a lower panel containing a table that will list your Ground Control Points (GCPs) as you add them.
3. To load your map, go to File, then Open Raster within the Georeferencer window. Browse to your Lec3/Data folder, select the downloaded Mount St. Helens map image, and click Open.

4. The scanned map will appear in the upper panel. Use the zoom and pan controls in the Georeferencer toolbar to familiarize yourself with the map's extent and content. Notice the printed grid lines with coordinate labels along the margins — these will serve as the foundation for our GCPs.

Step 2: Configuring Transformation Settings

Before placing any GCPs, you must define how QGIS will mathematically transform the image and what coordinate system the output should use.

Procedure

1. In the Georeferencer toolbar, click the gear icon to open the Transformation Settings dialog.
2. Set the Transformation Type to Polynomial 1 (Affine). This is a first-order linear transformation that corrects for translation, scaling, rotation, and shearing — the typical distortions introduced during scanning of a flat, professionally printed map. It requires a minimum of 3 GCPs. Higher-order transformations such as Polynomial 2 are generally not necessary for a high-quality archival scan like this USGS sheet, unless residuals remain unexpectedly high after placing your GCPs.
3. Set the Resampling Method to Nearest Neighbor. This method preserves the original pixel values without blending or averaging, which is appropriate for a scanned map where you want to maintain sharp linework and text.
4. For the Target CRS, click the Select CRS button. Since this is a USGS 15-minute topographic map from 1958, the coordinates printed on the map are in geographic latitude/longitude referenced to the NAD27 datum, which was the standard North American datum used by the USGS for most of the 20th century. Search for NAD27 and select NAD27 (EPSG:4267). Click OK.

Note on Datum Choice: Selecting the correct datum is critical. Although the visual difference between NAD27 and WGS84 may appear minor on screen, the coordinate offset between the two systems can exceed 50–100 meters in Washington State. For a map at 1:62500 scale, using the wrong datum will produce visible misalignment when you overlay the result on a modern basemap.

Step 3: Naming Your Output and Saving GCPs

Still within the Transformation Settings dialog:

1. For the output file, click the browse button next to the Output Raster field and navigate to your Lec3/Data folder. Name the output file `MtStHelens_1958_georef.tif`
2. Check the box labeled Save GCP Points. This saves your control points as a separate .points file alongside the output raster, allowing you to reload and refine them in future sessions without starting from scratch.
3. Ensure that Load in project when done is checked.
4. Click OK to confirm the transformation settings.

Step 4: Placing Ground Control Points (GCPs)

You will now identify locations in the scanned image whose real-world coordinates are known, and enter those coordinates manually.

Procedure

1. In the Georeferencer toolbar, click the Add Point button (crosshair icon).
2. Carefully examine the printed grid lines on the map. USGS 15-minute topographic maps display a geographic (latitude/longitude) graticule printed in black along the map margins. Locate a grid intersection — a point where a labeled latitude line crosses a labeled longitude line — near a corner of the map for your first GCP.
3. Click precisely on the intersection in the image. A dialog box will appear asking you to enter the map coordinates. Enter the X value (longitude — note that western longitudes are negative) and the Y value (latitude). Click OK.
4. The GCP table in the lower panel will now display your first entry.
5. Repeat this process, adding GCPs at additional grid intersections distributed across the entire map. Aim to cover all four corners and several interior points. For a Polynomial 1 transformation, you need a minimum of 3 GCPs, but more well-distributed points will yield a more accurate result.

Note on Coordinate Format: USGS maps often print coordinates in degrees, minutes, and seconds (e.g., 46°15'00" N, 122°15'00" W). You may also convert these to decimal degrees before entering them in the GCP dialog, but it is not mandatory. Do not forget the negative sign for western longitudes.

Step 5: Evaluating GCP Accuracy

Once you have entered at least 3 GCPs, the Georeferencer will automatically calculate error values for each point.

1. In the GCP table, observe the dX, dY, and Residual columns. These values report the discrepancy, in pixels, between where QGIS predicts a GCP should fall based on the fitted transformation and where you actually placed it.
2. A small, consistent residual across all points (ideally below 1–2 pixels for a well-scanned map) indicates a good fit.
3. If one point has a residual substantially higher than the others, it likely reflects a data-entry error in the coordinate values or an imprecise click on the grid intersection. You can correct this by double-clicking the Dest. X or Dest. Y value directly in the GCP table, or by deleting and re-capturing the point.
4. Distribute your GCPs evenly across the full extent of the image. GCPs clustered in one area leave other parts of the map poorly constrained, leading to increasing error toward the edges.

Step 6: Running the Georeferencing Process

Once you are satisfied with your GCP residuals:

1. Click the Start Georeferencing button (the play icon) in the Georeferencer toolbar.

2. QGIS will execute the polynomial transformation, warp the image to fit the target coordinate system, and write the output GeoTIFF to the location you specified in Step 3.
3. When the process is complete, the georeferenced raster (`MtStHelens_1958_georef.tif`) will be automatically loaded into your main QGIS canvas.

Step 7: Validating the Result with a Modern Basemap

This is where the historical significance of the map becomes visually apparent.

Procedure

1. Load the ESRI Topo basemap from the QuickMapServices plugin. (Web → QuickMapServices → ESRI → ESRI Topo)
2. In the Layers Panel, move the georeferenced 1958 map above the ESRI Topo layer so it renders on top.
3. Open the Layer Styling panel (the paintbrush icon in the Layers Panel) and navigate to the Transparency tab. Set the layer transparency of the 1958 map to approximately 40–50%. This allows the modern satellite imagery to show through, enabling direct visual comparison.
4. Zoom into the summit area of Mount St. Helens. On the 1958 map, the volcano appears as a nearly symmetrical, snow-capped cone rising to its pre-eruption elevation of approximately 2,950 meters (9,677 feet). The modern satellite imagery beneath it reveals the drastically different post-1980 landscape: a broad, horseshoe-shaped crater open to the north, a lava dome on the crater floor, and an extensive debris avalanche deposit filling the upper North Fork Toutle River valley.

The 1980 eruption of Mount St. Helens was the most significant volcanic event in the contiguous United States in the 20th century. The lateral blast on May 18, 1980 removed approximately 2.7 cubic kilometers of material from the summit and north flank in seconds, collapsing the mountain's elevation by roughly 400 meters and depositing debris across more than 600 square kilometers. Your georeferenced 1958 map provides a precise pre-eruption topographic baseline against which this transformation can be directly observed within a GIS environment.

Step 8: Refining Your Georeferencing (Optional)

If the map overlay appears misaligned with the satellite basemap in certain areas:

1. Reopen the Georeferencer tool via Layer, then Georeferencer.
2. Go to File, then Load GCP Points, and select the .points file saved alongside your output GeoTIFF. Your original GCPs will reload.
3. Load the original scanned map image again, add or adjust GCPs in the areas with the worst alignment, and re-run the georeferencing process to generate an improved output.

Summary

In this tutorial, you have:

1. Loaded a scanned historical raster into the QGIS Georeferencer and configured an appropriate transformation type and target CRS.
2. Placed Ground Control Points at known grid intersections and evaluated their accuracy using residual error values.
3. Exported a georeferenced GeoTIFF and saved your GCP file for future use.
4. Overlaid the 1958 topographic map against a modern ESRI satellite basemap to visually document the topographic impact of the 1980 Mount St. Helens eruption.

Georeferencing transforms static historical documents into analytically usable spatial data. The workflow you have practiced here is directly transferable to geological maps, historical land surveys, and any other scanned material where known coordinate information can be used to anchor the image in geographic space.

Homework: Visualizing the Eruption's Impact

Objective

Using your georeferenced 1958 topographic map as a reference, produce a map layout comparing the pre-eruption landscape of Mount St. Helens to its present-day appearance.

Procedure

1. Ensure your georeferenced `MtStHelens_1958_georef.tif` layer is loaded in QGIS alongside the ESRI Topo basemap. Set the transparency of the 1958 map layer to approximately 40-50%.
2. Navigate to Project, then New Print Layout. Name it Homework_3.
3. Set the page orientation to Landscape.
4. Use the Add Map tool to place a single map frame covering most of the page, centered on the Mount St. Helens summit area at a readable scale.
5. Add a title, scale bar, and north arrow.
6. Export the final layout as a PDF.

Submission Deliverable

Submit the exported single-page PDF map layout showing the georeferenced 1958 topographic map overlaid on the modern ESRI Topo basemap.